Effora B2C — RTO Analysis & Recovery Opportunity

Domain: E-commerce Logistics | Tools: Excel (Pivot Tables) | Data: 4,045 Shipments, Nov–Apr

Summary

Objective

Diagnose why Return-to-Origin (RTO) shipments were eating into Effora's e-commerce revenue, and identify where operational or payment-related fixes could recover the most blocked order value.

Concepts / Methods Used

Pivot Table Analysis (State-wise, Region-wise, Date-wise success rates), Cross-tabulation (Payment Type vs RTO outcome), Root-cause segmentation (Invalid Address, Attempt Count, TAT Breach Reason).

Key Output

- Overall RTO rate: 15.33% across 4,045 shipments (620 RTO orders)
- COD orders RTO at 21.48% vs Pre-paid orders at just 0.17% — a ~125x difference
- ₹9.69 lakh in order value blocked in RTO shipments; 99.7% of that (₹9.66 lakh) is tied to COD orders alone
- 56.6% of RTO orders had a Valid Address on file — address quality is not the primary RTO driver
- 72.3% of RTO shipments were within TAT (no delivery delay) — most RTOs are not a logistics-speed problem
- West and South regions carry the highest RTO volume (194 and 273 cases) though not the highest RTO rate
- East region has the weakest success rate (67.03%) and highest RTO% (26.49%) among all zones

Business Actionables

- Prioritize COD-to-prepaid conversion incentives — this single lever addresses the majority of blocked value
- Do not over-invest in address-verification fixes — most RTOs occur even with valid addresses on file
- Introduce pre-delivery COD confirmation (call/WhatsApp) specifically for East region and low-success states
- Use state-wise success-rate ranking to prioritize courier/ops review in the weakest-performing states first

Detailed Analysis — Actual Excel Output, Explained, With Actionables

Output 1 — State-Wise Success & RTO Pivot Table

	Data			Success %	RTO%
D State	Count of Order_ID	Sum of Success Count	Sum of RTO Count	#VALUE!	#VALUE!
Andhra Pradesh	186	149	32	80.1	17.2
Arunachal Pradesh	2	1	0	50.0	0.0
Assam	41	23	17	56.1	41.5
Bihar	21	14	4	66.7	19.0
Chandigarh	9	7	1	77.8	11.1
Chhattisgarh	26	20	6	76.9	23.1
Dadra and Nagar Haveli	1	1	0	100.0	0.0
Daman & Diu	1	1	0	100.0	0.0
Delhi	154	123	29	79.9	18.8
Goa	44	35	8	79.5	18.2
Gujarat	227	198	24	87.2	10.6
Haryana	94	76	13	80.9	13.8
Himachal Pradesh	9	4	4	44.4	44.4
Jammu & Kashmir	8	4	2	50.0	25.0
Jharkhand	14	8	6	57.1	42.9
Karnataka	566	468	93	82.7	16.4
Kerala	130	110	17	84.6	13.1
Ladakh	2	2	0	100.0	0.0
Madhya Pradesh	70	57	10	81.4	14.3
Maharashtra	1035	864	146	83.5	14.1
Meghalaya	1	1	0	100.0	0.0
Mizoram	5	5	0	100.0	0.0
Nagaland	2	1	1	50.0	50.0
Orissa	31	19	9	61.3	29.0
Pondicherry	19	13	5	68.4	26.3
Punjab	89	70	16	78.7	18.0
Rajasthan	67	49	14	73.1	20.9
Sikkim	1	0	0	0.0	0.0
Tamil Nadu	700	608	85	86.9	12.1
Telangana	268	221	41	82.5	15.3
Tripura	4	2	1	50.0	25.0
Uttar Pradesh	138	107	26	77.5	18.8
Uttarakhand	18	17	1	94.4	5.6
West Bengal	62	49	11	**79.0322580645161**	17.7

Actual Excel pivot table: Count of Orders, Success Count, RTO Count by State

What this is: A pivot table built from the Consolidated Data sheet, breaking down every shipment by destination state, with total orders, successful deliveries, RTO count, success %, and RTO % computed per state.

Layman meaning: Delivery success is uneven across India. States like Gujarat (87.2% success) and Kerala (84.6%) perform well, while Assam (56.1%), Jharkhand (57.1%), and Himachal Pradesh (44.4%) show much weaker outcomes with RTO rates as high as 41–44%. This isn't a single national problem — it's concentrated in specific geographies.

Actionable for the user: Instead of a blanket fix, Effora can target operational review (courier partner, delivery process, COD verification) specifically in the handful of low-performing states, which is a far more efficient use of limited resources than a national-level intervention.

Output 2 — Region-Wise Success & RTO Pivot Table

	Data			Success %	RTO%
Region	Count of Order_ID	Sum of Success Count	Sum of RTO Count	#VALUE!	#VALUE!
E	185	124	49	67.03	26.49
N	588	459	106	78.06	18.03
W	1403	1175	194	83.75	13.83
S	1869	1569	273	**83.95**	**14.61**

Actual Excel pivot table: Region-wise Order Count, Success Count, RTO Count

What this is: The state-level data rolled up into four national zones — East (E), North (N), West (W), South (S) — with aggregate success and RTO rates for each.

Layman meaning: East is the weakest zone by rate: only 67.03% success and 26.49% RTO, clearly behind North (78.06%), West (83.75%), and South (83.95%). West and South carry the largest shipment volumes (1,403 and 1,869 orders respectively), so while their RTO rate is lower, their absolute RTO case count is still substantial (194 and 273 cases).

Actionable for the user: Two different strategies are needed: in East, the priority is fixing the rate itself (root-cause diagnosis); in West and South, the priority is absolute case-count reduction given the sheer shipment volume — even a small rate improvement there recovers more total value.

Output 3 — Growth Opportunities: State Ranking by Success %

State	Sum of Success Count	Success %
Maharashtra	864	83.5
Tamil Nadu	608	86.9
Karnataka	468	82.7
Telangana	221	82.5
Gujarat	198	87.2
Andhra Pradesh	149	80.1
Delhi	123	79.9
Kerala	110	84.6
Uttar Pradesh	107	77.5
Haryana	76	80.9
Punjab	70	78.7
Madhya Pradesh	57	81.4
West Bengal	49	79.0
Goa	35	79.5
Chhattisgarh	20	76.9
Uttarakhand	17	94.4
Chandigarh	7	77.8
Mizoram	5	100.0
Ladakh	2	100.0
Dadra and Nagar Haveli	1	100.0
Daman & Diu	1	100.0
Meghalaya	1	100.0

Actual Excel pivot table: States ranked by delivery success rate

What this is: A ranked view of every state by success percentage, isolating which states are already strong (candidates for scaling shipment volume) versus which are weak (candidates for intervention before scaling).

Layman meaning: High-volume, high-success states like Maharashtra (83.5%) and Tamil Nadu (86.9%) are proven markets — safe to grow further. Low-success states such as Bihar (66.7%), Assam (56.1%), and Jharkhand (57.1%) are growth traps: pushing more marketing spend into these states without fixing delivery would just generate more RTO.

Actionable for the user: Effora's marketing/growth team should sequence expansion spend by this ranking — scale in green states, pause or fix-first in red states. This directly connects a logistics analysis to a marketing budget decision.

Output 4 — COD vs Prepaid: RTO Rate and Blocked Order Value

Payment Type	Total Orders	RTO Count	RTO %	Blocked Order Value (INR)
COD	2877	618	21.48%	965,840.20
Pre-paid	1168	2	0.17%	2,907.20
Grand Total	**4045**	**620**	**15.33%**	**968,747.40**

Excel summary table built from Consolidated Data (Payment Type × Current Status cross-tabulation)

What this is: A summary comparing RTO outcomes and blocked order value between COD (Cash on Delivery) and Pre-paid orders, derived from the full 4,045-row shipment dataset.

Layman meaning: The gap is stark: COD orders RTO at 21.48% while Pre-paid orders RTO at only 0.17% — COD shipments are roughly 125 times more likely to come back undelivered. In money terms, ₹9.69 lakh worth of orders is currently stuck in RTO, and 99.7% of that value (₹9.66 lakh) sits in COD orders. Pre-paid RTO is close to a non-issue (₹2,907 only).

Actionable for the user: This is the single highest-leverage fix available: any initiative that shifts customers from COD to prepaid — small prepaid discounts, easier UPI checkout, COD convenience fees — will recover far more blocked value than any address-quality or logistics-speed fix, because those are not where the problem actually lives (see Output 5).

Output 5 — RTO Root-Cause Breakdown (from Consolidated Data)

What this is: A breakdown of the 620 RTO orders by address validity, delivery attempt count, and TAT (Turn-Around-Time) breach status, computed directly from the Consolidated Data sheet's tracking fields.

Layman meaning: 351 of 620 RTO orders (56.6%) had a Valid Address on file — meaning the address was not the reason for failure in the majority of cases. 328 orders (52.9%) went to RTO after just a single delivery attempt, and 448 orders (72.3%) were within TAT, meaning the courier was not late. This combination points away from "bad address" or "slow logistics" as the main driver, and toward customer-side refusal or non-response — a pattern strongly associated with COD orders, where the customer has not pre-paid and has no financial commitment to accept the parcel.

Actionable for the user: Don't spend budget on address-verification tools or courier-speed upgrades expecting an RTO fix — the data shows most RTOs happen on valid addresses within TAT, so the fix has to target payment behavior and delivery-attempt conversion (e.g. pre-delivery confirmation calls before the first attempt), not logistics infrastructure.

Business Takeaway

Effora's RTO problem is a payment-behavior problem, not a logistics or address-quality problem. COD orders drive over 99% of blocked order value despite being roughly 71% of shipment volume, and RTOs occur even on valid addresses within TAT. The highest-leverage fix is COD-to-prepaid conversion, sequenced with state-level prioritization using the success-rate ranking already built.